

Mark schemes

Q1.

- (a) Ball is a particle/no spin

B1: One assumption

B1

No air resistance/Only gravity or weight

B1: Second assumption

B1

2

- (b) (i) $24.5 t - 4.9 t^2 = 0$

M1: Equation for vertical motion with height zero

M1

$$\left(t = 0 \text{ or } t = \frac{24.5}{4.9} = 5s \right)$$

AG

A1: Correct equation

A1

dM1: Solving for t

dM1

A1: Correct time from correct working

A1

4

- (b) (ii) $R = 10 \times 5 = 50 \text{ m}$

M1: Use of horizontal component of velocity to find the range

M1

A1: Correct range

A1

2

- (c) $20 = 10 t$

M1: Horizontal equation

M1

$$t = 2$$

A1: Time to reach wall

A1

$$h = 24.5 \times 2 - 4.9 \times 2^2 = 29.4 \text{ m}$$

*dM1: Vertical equation for height with
u = 24.5 and a negative acceleration*

dM1

A1: Correct height

A1

4

- (d) No change as acceleration and initial velocity do not change with the mass

B1: No change

B1

B1: Explanation

B1

2

[14]

Q2.

(a) $\mathbf{a} = 8 \cos t \mathbf{i} - 4 \sin t \mathbf{j}$

M1 differentiation

M1A1

$$\mathbf{F} = 2 \cos t \mathbf{i} - \sin t \mathbf{j}$$

A1

3

(b) (i) $|\mathbf{F}| = \sqrt{(4 \cos^2 t + \sin^2 t)}$

Magnitude

M1

$$= \sqrt{(3 \cos^2 t + 1)}$$

m1

CAO

A1

3

(ii) $|\leq \mathbf{F}| \leq 4$

B1 each value (at end of range)

B1B1

2

[8]

Q3.

(a) (i) $\mathbf{v} = 2 \cos 2t \mathbf{i} + 6 \mathbf{j}$

M1 differentiation (6t)

M1A1

2

(ii) $|\mathbf{v}| = \sqrt{4 \cos^2 2t + 36}$

Sum of squares, for v or v²

M1

ft trig term v

A1F

CAO

A1

3

(iii) $\cos^2 2t = 0$ or $\cos 2t = 0$

M1

$t = \frac{\pi}{4}$

radians

A1

2

(b) (i) $\mathbf{a} = -4 \sin 2t \mathbf{i}$

Differentiation attempt

M1

$\mathbf{F} = 0.25 \mathbf{a}$

Used

M1

$\mathbf{F} = -\sin 2t \mathbf{i}$

ft v, see vector

A1F

3

(ii) Direction is $\pm \mathbf{i}$

B1

$$|\sin 2t| \leq 1$$

B1

2

[12]

Q4.

Answers in (a), (b) and (c)(i) must be vectors

(a) $\mathbf{a} = 4t\mathbf{i} + 3\mathbf{j}$

Must be vector, both i and j present

B1

1

(b) $\mathbf{v} = 2t^2\mathbf{i} + 3t\mathbf{j}$

Integration of acceleration attempted

M1

ft a, both i and j present

A1F

2

(c) (i) $\mathbf{r} = \int_0^t (2t^2\mathbf{i} + 3t\mathbf{j}) dt$

Integration attempted

M1

$$= \frac{2t^3}{3}\mathbf{i} + \frac{3t^2}{2}\mathbf{j}$$

*A1 each term. Condone i and j missing.
 -1 for constants present,
 -1 for both terms unsimplified*

A1A1

3

(ii) $\left. \begin{array}{l} \mathbf{v} \text{ parallel to } \mathbf{i} + \mathbf{j} \\ 2t^2 = 3t \end{array} \right\}$

M1

$$t = \frac{3}{2}$$

Solve, ft v with vector components

A1F

$$\mathbf{r} = \frac{9}{4}\mathbf{i} + \frac{27}{8}\mathbf{j}$$

A1F

$$\text{Distance} = \sqrt{\left(\frac{9}{4}\right)^2 + \left(\frac{27}{8}\right)^2}$$

Magnitude of $\mathbf{r}$ in any form

M1

$$= 4.06 \text{ (4.056)}$$

Numerical answer, ft candidate's $\mathbf{r}$ and t

A1F

5

[11]

Q5.

(a) (i) $x = 14t$

Accept as components of position vector

Accept $x = 15.7 \cos 26.6$

B1

$$y = 7t - 4.9t^2$$

Use of equation

M1

3

Substitution

A1

$$\text{Accept } g; \text{ accept } y = 15.7 \sin 26.6t - \frac{1}{2}gt^2$$

(ii) $t = \frac{x}{14} \quad y = 7 \times \frac{x}{14} - 4.9 \times \left(\frac{x}{14}\right)^2$

Use of candidate's coordinates

M1

Substituted; accept unsimplified

A1

$$= \frac{x}{2} - 4.9 \frac{x^2}{196}$$

$$y = \frac{20x - x^2}{40}$$

AG; convincingly found

A1

3

(iii) $y = 0$ $20x - x^2 = 0$

Full method for range ($t = 1.4286$ for range,
or $t = 0.7143$ for greatest height and horizontal

distance then doubled, or $R = \frac{V^2 \sin 2\alpha}{g}$)

M1

$x = 20$

AWRT

A1

2

(b) $x = 12, y = 2.4$

Substitution into appropriate equation(s), ($t = 0.857$)

M1

For $y = 2.4$

A1

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ft provided $y > 0$

Alt: M1 for $x = 10$ subs and followed by comparison,
 $H = 2.5$ A1, conclusion A1

A1F

3

(c) Air resistance ignored, ball treated as particle...

B1

1

[12]

Q6.

(a) $x = 7t$

B1

$$y = 0 + \frac{1}{2}gt^2$$

Accept ±

M1

$$y = 4.9t^2$$

Accept ±

A1

3

(b) $t = \frac{x}{7}$

Attempt at substitution, or use of equation of trajectory with $V = 7$ & $\alpha = 0$

M1

$$y = 4.9 \left(\frac{x}{7} \right)^2$$

$$y = \frac{x^2}{10}$$

CAO

A1

2

(c) $8.1 = \frac{x^2}{10}$

Full method for x , accept ±

M1

$$x = 9\text{m}$$

AWRT 9.0 if two stages used

A1

2

(d) vert: $v^2 = 0 + 2 \times 9.8 \times 8.1$

*For M1 Accept ±,
for A1 consistency of signs needed*

M1A1

$$u = 7$$

B1

$$\text{speed}^2 = (7^2 + 12.6^2)$$

M1

$$\text{speed} = 14.4 \text{ ms}^{-1}$$

$$(14.414)$$

A1F

5

[12]

Q7.

(a) $\mathbf{v} = 6\mathbf{i} + 2t\mathbf{j}$

M1: differentiation attempted and vector quantity for $\mathbf{v}$ given

M1A1

2

(b) $\text{sp} = \sqrt{6^2 + 4t^2} \text{ ms}^{-1}$

M1: sum of squares attempted giving scalar expression

*A1 : all correct, accept $(2t)^2$
f.t. $\mathbf{v}$ with 2 components*

M1A1F

2

(c) $\sqrt{6^2 + 4t^2} = 6\sqrt{2}$

o.e.; scalar expression for $\mathbf{v}$ in terms of t from (b) used

M1

$$36 + 4t^2 = 36 \times 2$$

$$t = 3$$

f.t. minor slip in (b) provided t is positive solution of quadratic eqn

A1F

2

[6]

Q8.

(a) $\mathbf{r} = \int (4\mathbf{i} - 2t\mathbf{j}) \, dt$

attempted

M1

$$= 4ti - t^2j \quad (+c)$$

A1

$$t = 0, \mathbf{r} = 8\mathbf{j}$$

used

m1

$$\mathbf{r} = 4ti + (8i - t^2)\mathbf{j}$$

A1F

4

(b) $t = 2, \mathbf{r} = 8\mathbf{i} + 4\mathbf{j}$

B1F

1

(c) $8 - t^2 = 0$

M1

$$t = 2\sqrt{2} \quad \text{or} \quad t = 2.83$$

A1

2

[7]

Q9.

(a) (i) vertical: $s = ut + \frac{1}{2}at^2$
 $0 = 7t - 4.9t^2$

Full method

M1

$$0 = t(7 - 4.9t)$$

$$0 = 7 - 4.9t$$

m1

$$t = \frac{10}{7} \text{ sec (1.43)}$$

A1

3

(ii) $OF = 21 \times \frac{10}{7}$

M1

$$= 30 \text{ m}$$

A1F

2

(b) vert : $0 = 3.5t - 4.9t^2$

Full method required

M1

$$t = \frac{3.5}{4.9}$$

m1

$$FG = 21 \times \frac{3.5}{4.9}$$

M1

$$= 15 \text{ m}$$

A1

4

(c) $GH = \frac{1}{2} \times 15 = 7.5$

B1F

$$OH = 30 + 15 + 7.5$$

M1

$$= 52.5 \text{ m}$$

A1F

3

[12]

Q10.

(a) $\Gamma = \int (2t\mathbf{i} + 4t\mathbf{j})dt$

Integration attempt for M1

M1

$$= t^2\mathbf{i} + 4t\mathbf{j} (+c)$$

A1

$$t = 0, \Gamma = 2\mathbf{j}, 2\mathbf{j} = \mathbf{c}$$

m1

$$\Gamma = t^2\mathbf{i} + (4t + 2)\mathbf{j}$$

Any vector form

A1F

4

(b) $t = 2, \Gamma = 4\mathbf{i} + 10\mathbf{j}$

subs either value of t into $\mathbf{r}$ for M1

M1 A1F

$$t = 4, \Gamma = 16\mathbf{i} + 18\mathbf{j}$$

ft for vectors with 2 non-zero components

A1F

$$= 12\mathbf{i} + 8\mathbf{j}$$

subtraction of two vectors

A1F

4

Alternative:

$$[t^2\mathbf{i} + 4t\mathbf{j}]^4$$

M1

$$= (16\mathbf{i} + 16\mathbf{j}) - (4\mathbf{i} + 8\mathbf{j})$$

A1F A1F

$$= 12\mathbf{i} + 8\mathbf{j}$$

A1F

subtraction

[8]

Q11.

(a) $x = 7t, y = 7t - \frac{1}{2} \times 9.8t^2$ $\begin{pmatrix} 7t \\ 7t - 4.9t^2 \end{pmatrix}$

B1 horiz

M1 use of $s = ut + \frac{1}{2}at^2$

M1A1 for vert, accept 10 or g

B1M1A1

3

(b) $t = \frac{x}{7}$

B1

$$y = 7 \times \frac{x}{7} - \frac{1}{2} \times 9.8 \times \frac{x^2}{49}$$

M1A1

$$y = x - 0.1x^2$$

A1

4

Alternative (full verification method)

$x = 7t$ used B1

subs into $y = x - 0.1x^2$

M1A1

$y = 7t - 4.9t^2$ as in (a)

A1

Alternative: (use equation of path)

$$y = x \tan \alpha - \frac{gx^2}{2v^2 \cos^2 \alpha}$$

45° used

B1

subs

M1A1

simplify

A1

(c) $x = 4, \quad y = 4 - 0.1 \times 4^2$

or $t = \frac{4}{7} (0.571(43))$

M1

$y = 2.4$

A1

above wire as $2.4 > 2$

ft slip in y provided y positive

A1F

3

(d) $y = 0$
 $x - 0.1x^2 = 0$

M1

$$x(1 - 0.1x) = 0$$

m1 A1

$$x = 10$$

A1F

4

Alternative: (use of coordinate equations)

$$y = 0, 7t - 4.9t^2 = 0$$

M1

$$\Rightarrow t = \frac{10}{7} \quad (1.43)$$

m1A1

$$x = 7t = 10$$

A1F

(Accept 10.01 if $t = 1.43$ used)

Alternative: (use of symmetry)

$$v = 0, 0 = 7 - 9.8t$$

M1

$$t = \frac{5}{9.8}$$

A1

$$x = 7 \times t \times 2$$

m1

$$x = 10 \text{ (acc 9.996)}$$

A1F

Alternative: (use of equation for Range)

$$R = \frac{2V^2 \sin \alpha \cos \alpha}{g}$$

Correct formula used M1, 45° used m1

Correct subs A1,

result A1F, AWRT 10m, ft one slip.

(e) $\text{Speed}^2 = 7^2 + 7^2$

M1

$= 9.90 \text{ ms}^{-1}$

Accept 9.9, $7\sqrt{2}$,
or 9.91 if $t = 1.43$ used in downward motion

A1

2

[16]

Q12.

(a) $\mathbf{v} = (2t - 6)\mathbf{i} + t^2\mathbf{j}$

differentiation

M1 A1, A1

3

(b) $2t - 6 = 0$

M1

$t = 3$

ft differentiation for $t > 0$

A1F

2

(c) $\mathbf{a} = 2\mathbf{i} + 2t\mathbf{j}$

M1

$\mathbf{a}$ is function of t , not constant

Differentiation of 2 components of vector
expression for $\mathbf{v}$ attempted

CAO $\mathbf{a}$ correct and explanation correct;
reference to dependence on t required

A1

2

[7]

Q13.

(a) Vertical: $0 = 4Ut - \frac{1}{2}gt^2$

Accept use of 9.8

M1A1

$t = \frac{8U}{g}$

A1

3

(b) horizontal: $OH = 3U \times \frac{8U}{g}$
Use of $v \times t$

M1

$$OH = \frac{24U^2}{g}$$

A1

2

(c) $\text{speed}^2 = (3U)^2 + (4U)^2$

M1

$$\text{speed} = 5U$$

A1

2

(d) $(\sqrt{18}U)^2 = (3U)^2 + v^2$

M1A1

$$v = (\pm)3U$$

A1

$$\pm 3U = 4U - gt$$

Or attempt to solve quadratic in t

M1

$$t_1 = \frac{U}{g} \quad t_2 = \frac{7U}{g}$$

A1A1

6

[13]

Q14.

(a) $-16i + 16j = \frac{1}{2} \mathbf{a} \times 4^2$

Use of vector constant acceleration equation

M1

Correct equation

A1

$$\mathbf{a} = \frac{-16\mathbf{i} + 16\mathbf{j}}{8} = -2\mathbf{i} + 2\mathbf{j}$$

Solving for a

M1

ag Correct a

A1

4

(b) $\mathbf{v} = 5(-2\mathbf{i} + 2\mathbf{j}) = -10\mathbf{i} + 10\mathbf{j}$

Expression for v at t = 5

M1

Correct expression

A1

$$v = \sqrt{10^2 + 10^2} = 14.1 \text{ ms}^{-1} \text{ (to 3 sf)}$$

Finding magnitude

M1

Correct speed

A1

4

(c) $20\mathbf{i} - 10\mathbf{j} + \mathbf{Q} = 3(-2\mathbf{i} + 2\mathbf{j})$

Application of Newton's second law to form equation

M1

Correct equation

A1

$$\mathbf{Q} = (-20 - 6)\mathbf{i} + (10 + 6)\mathbf{j} = -26\mathbf{i} + 16\mathbf{j}$$

Solving for Q

M1

Correct Q

A1

4

[12]

Q15.

- (a) $\mathbf{v} = -4e^{-t} \mathbf{i} + (6 - 3e^{-t})\mathbf{j}$
Differentiating position vector

M1

$$t = 0 \text{ A1}$$

Correct velocity

$$\mathbf{v} = -4\mathbf{i} + 3\mathbf{j}$$

ag Substituting $t = 0$ to obtain initial velocity

A1

3

- (b) $\mathbf{a} = 4e^{-t} \mathbf{i} + 3e^{-t} \mathbf{j}$
Differentiating velocity

M1

Correct acceleration

A1

2

- (c) $\mathbf{a} = 4\mathbf{i} + 3\mathbf{j}$
Finding acceleration when $t = 0$

M1

$$a = \sqrt{4^2 + 3^2} = 5$$

Correct magnitude

A1

2

- (d) $\mathbf{v} \rightarrow 0\mathbf{i} + 6\mathbf{j}$
For $\mathbf{i}$ component

B1

For $\mathbf{j}$ component

B1

2

[9]

Q16.

- (a) $\mathbf{v} = (3\mathbf{i} - 10\mathbf{j}) + (4\mathbf{i} + 2\mathbf{j})t$
Use of $\mathbf{v} = \mathbf{u} + \mathbf{a}t$

M1

$$= (3 + 4t) \mathbf{i} + (2t - 10) \mathbf{j}$$

ag Correct result from correct working

A1

2

(b) $2t - 10 = 0$

j component equal to zero

M1

$t = 5$

Correct time

A1

2

(c) (i) $\mathbf{r} = (3\mathbf{i} - 10\mathbf{j}) \times 10 + \frac{1}{2} (4\mathbf{i} + 2\mathbf{j}) \times 10^2$
Finding r when t = 10

Correct expression

M1

$= 230\mathbf{i}$

Correct final answer

A1

A1

3

(c) (ii) $\mathbf{v} = (3 + 4 \times 10)\mathbf{i} + (2 \times 10 - 10)\mathbf{j}$
 $= 43\mathbf{i} + 10\mathbf{j}$

Correct velocity

B1

1

(d) $\mathbf{r} = 230\mathbf{i} + (43\mathbf{i} + 10\mathbf{j}) \times 10$

Uses zero acceleration

M1

Uses both answers from (c)

M1

$= 660\mathbf{i} + 100\mathbf{j}$

Correct expression for *r*

A1

Correct simplified result

A1

$$r = \sqrt{660^2 + 100^2}$$

Finding magnitude

M1

$$= 668 \text{ m}$$

Correct distance

Follow through from part (c)

A1 ✓

6

[14]

Q17.

(a) $5\mathbf{i} - 2\mathbf{j} = 4\mathbf{i} + 3\mathbf{j} + 10\mathbf{a}$

Use of constant acceleration equation for velocity

M1

Correct equation

A1

$$\mathbf{a} = \frac{1}{10} (\mathbf{i} - 5\mathbf{j}) = (0.1\mathbf{i} - 0.5\mathbf{j}) \text{ ms}^{-2}$$

ag *Correct acceleration from correct working*

A1

3

(b) $\mathbf{r} = (4\mathbf{i} + 3\mathbf{j})t + 0.5(0.1\mathbf{i} - 0.5\mathbf{j})t^2$

Use of constant acceleration equation for position

M1

Correct expression

A1

2

(c) $\mathbf{r} = (4t + 0.05t^2)\mathbf{i} + (3t - 0.25t^2)\mathbf{j}$

j component is zero

M1

Correct equation

A1

$$3t - 0.25t^2 = 0$$

$$t(3 - 0.25t) = 0$$

Solving for t

m1

$$t = 0 \text{ or } t = \frac{3}{0.25} = 12 \text{ s}$$

$$t = 12 \text{ s}$$

Correct time

A1

4

(d) $\mathbf{v} = (4 + 0.1t)\mathbf{i} + (3 - 0.5t)\mathbf{j}$

$$3 - 0.5t = 0$$

$$t = 6$$

Expression for velocity

M1

Correct velocity

A1

j component is zero

m1

Correct time

A1

4

[13]

Q18.

(a) $60\mathbf{i} + 20\mathbf{j} = 20(2\mathbf{i} + 3\mathbf{j}) + 200\mathbf{a}$

$$20\mathbf{i} + 80\mathbf{j} = 200\mathbf{a}$$

$$\mathbf{a} = 0.1\mathbf{i} + 0.4\mathbf{j}$$

*Use of constant acceleration equation
in vector form to find a*

M1

Correct equation

A1

Correct a

A1

3

(b) $\mathbf{v} = (2\mathbf{i} - 3\mathbf{j}) + (0.1\mathbf{i} + 0.4\mathbf{j})t$

Use of $\mathbf{v} = \mathbf{u} + \mathbf{at}$

M1

$$= (2 + 0.1t)\mathbf{i} + (-3 + 0.4t)\mathbf{j}$$

Correct expression

A1

2

(c) $2 + 0.1t = -(-3 + 0.4t)$

Equating components with $\pm$

M1

$$0.5t = 1$$

Correct equation

A1

$$t = 2$$

A1

$$\mathbf{v} = 2.2\mathbf{i} - 2.2\mathbf{j}$$

Finding velocity

M1

$$v = \sqrt{2.2^2 + 2.2^2} = 3.11\text{ms}^{-1}$$

Correct speed

A1

5

EXAM PAPERS PRACTICE

[10]

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Q19.

(a) $19\mathbf{i} - 25\mathbf{j} = \frac{1}{2} \mathbf{a} \times 10^2 + 9\mathbf{i} + 10\mathbf{j}$

Using both position vectors to form a constant acceleration equation

M1

$$50\mathbf{a} = 10\mathbf{i} - 35\mathbf{j}$$

Correct equation
Solving for $\mathbf{a}$

A1 M1

$$\mathbf{a} = 0.2\mathbf{i} - 0.7\mathbf{j}$$

Correct $\mathbf{a}$

A1

4

(b) $\mathbf{v} = 10(0.2\mathbf{i} - 0.7\mathbf{j})$
use of $\mathbf{v} = \mathbf{at}$

M1

$= 2\mathbf{i} - 7\mathbf{j}$
Correct velocity

A1

$v = \sqrt{2^2 + 7^2} = 7.28 \text{ ms}^{-1}$
Finding speed from velocity
Correct speed

m1 A1

4

(c) $15.4 = \frac{1}{2} \times 0.2 \times t^2 + 9$
Finding t from one component

M1

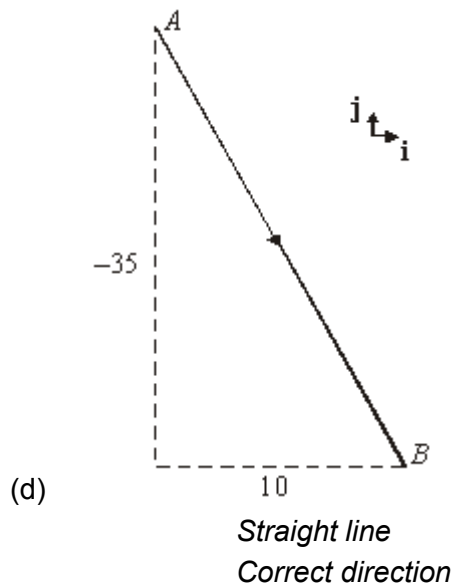
$t = 8$
Correct t

A1

$\frac{1}{2} \times (-0.7) \times 8^2 + 10 = -12.4$
Using $t = 8$ with other component
Correct result

M1 A

4



B1 B1

2

[14]

Q20.

(a) $40\mathbf{i} - 15\mathbf{j} = \frac{1}{2} (5\mathbf{i} - 2\mathbf{j} + \mathbf{v}) \times 10$

Use of constant acceleration equation with $\mathbf{v}$ unknown
Correct equation

M1 A1

$$\mathbf{v} = 8\mathbf{i} - 3\mathbf{j} - 5\mathbf{i} + 2\mathbf{j} = 3\mathbf{i} - \mathbf{j}$$

Solving for $\mathbf{v}$
Correct $\mathbf{v}$

M1 A1

4

(b) $3\mathbf{i} - \mathbf{j} = 5\mathbf{i} - 2\mathbf{j} + 10\mathbf{a}$

Use of $\mathbf{v} = \mathbf{u} + \mathbf{at}$

M1A1

$$\mathbf{a} = -0.2\mathbf{i} + 0.1\mathbf{j}$$

Correct $\mathbf{a}$

A1

3

(c) $\mathbf{F} = 15(-0.2\mathbf{i} + 0.1\mathbf{j})$

Use of $\mathbf{F} = m\mathbf{a}$

M1

$$= -3\mathbf{i} + 1.5\mathbf{j}$$

Correct **F**

A1

$$F = \sqrt{3^2 + 1.5^2} = 3.35$$

Finding magnitude of **F**

Correct magnitude

M1 A1

4

[11]

Q21.

(a) (i) $\mathbf{a} = \frac{1}{4} (8\mathbf{i} - 3\mathbf{j}) = 2\mathbf{i} - 3\mathbf{j}$

Use of $\mathbf{F} = m\mathbf{a}$, must be applied to both components

M1

Correct acceleration

A1

2

(ii) $\mathbf{v} = 20(2\mathbf{i} - 3\mathbf{j}) = 40\mathbf{i} - 60\mathbf{j}$

Use of $\mathbf{v} = \mathbf{u} + \mathbf{at}$ with $\mathbf{u} = 0\mathbf{i} + 0\mathbf{j}$

M1

Correct **v**

A1

2

(iii) $\mathbf{r} = \frac{1}{2} (2\mathbf{i} - 3\mathbf{j}) \times 20^2$

Use of constant acceleration equation to find **r** with $\mathbf{u} = 0\mathbf{i} + 0\mathbf{j}$

M1

Correct expression

A1

$$= 400\mathbf{i} - 600\mathbf{j}$$

Correct **r** in simplified form

A1

3

(b) $\mathbf{r} = 400\mathbf{i} - 600\mathbf{j} + 25(40\mathbf{i} - 60\mathbf{j})$

Use of $r + 25v$

M1

Correct expression

A1ft

$$= 1400\mathbf{i} - 2100\mathbf{j}$$

Correct position vector

A1ft

$$r = \sqrt{1400^2 + 2100^2} = 2520 \text{ m (to 3 sf)}$$

Finding magnitude

m1

Correct distance from correct working

A1ft

5

Alternative: Straight line method

M1 two distances r and $25v$

A1 for each distance

m1 adding

A1 correct final answer

[12]

Q22.

(a) $\mathbf{r} = 5\mathbf{j}$

Correct vector

B1

1

(b) $4t - 0.01t^2 = 0$

Equation based on i component

M1

Correct equation

A1

$$t = 0 \text{ or } t = 400$$

Solving the quadratic

M1

$$t = 400$$

Selecting $t = 400$

A1

(c) $\mathbf{v} = (4 - 0.02t)\mathbf{i} + (-3 - 0.08t)\mathbf{j}$

Use of $\mathbf{v} = \mathbf{u} + \mathbf{at}$

M1

Correct velocity vector

A1

$$4 - 0.02t = 3 + 0.08t$$

Equation using both components

M1

Correct equation

A1

$$t = 10$$

Correct t

A1

5

[10]

Q23.

(a) $19\mathbf{i} + 13\mathbf{j} = 35\mathbf{i} + 45\mathbf{j} + 8\mathbf{a}$

Constant acceleration equation to find $\mathbf{a}$

M1

$$\mathbf{a} = \frac{19-35}{8}\mathbf{i} + \frac{13-45}{8}\mathbf{j} = 2\mathbf{i} - 4\mathbf{j}$$

Correct equation

A1

ag *Correct $\mathbf{a}$ from correct working*

A1

3

(b) $\mathbf{r} = (35\mathbf{i} + 45\mathbf{j})t + \frac{1}{2}(-2\mathbf{i} - 4\mathbf{j})t^2$

Use of constant acceleration equation

M1

Correct $\mathbf{i}$ component

A1

Correct $\mathbf{j}$ component

A1

3

(c) $\mathbf{r} = (3.5t - t^2)\mathbf{i} + (45t - 2t^2)\mathbf{j}$

Splitting into components correctly

B1

$$35t - t^2 = 300$$

Forming equation for one component

M1

$$t^2 - 35t + 300 = 0$$

Solving quadratic

M1

$$t = 15 \text{ or } t = 20$$

Two correct solutions

A1

$$4t - 2t^2 = 225$$

Forming correct second quadratic

B1

$$2t^2 - 45t + 225 = 0$$

$$t = 7.5 \text{ or } t = 15$$

Solving quadratic for two solutions

M1

$$t = 15 \text{ seconds}$$

Correct final solution

A1

7

[13]

Q24.

(a) $\mathbf{F} = (3\mathbf{i} + 4\mathbf{j}) + (6\mathbf{i} - 8\mathbf{j})$

Addition of the two forces

M1

2

$$= 9\mathbf{i} - 4\mathbf{j}$$

Correct resultant

A1

(b) $F = \sqrt{9^2 + 4^2} = 9.85 \text{ N}$

Finding magnitude

M1

Correct magnitude

A1

2

(c) $\tan \alpha = \frac{4}{9}$

Using tan to find the angle

M1

Correct equation

A1

$\alpha = 24.0^\circ$

Correct angle

A1

3

[7]

Q25.

(a) $2\mathbf{i} + 14\mathbf{j} = \mathbf{u} + 2(2\mathbf{i} - 3\mathbf{j})$

Using constant acceleration equation to find $\mathbf{u}$

Correct equation

M1

Correct $\mathbf{u}$ from correct working

A1

$\mathbf{u} = (2 - 4)\mathbf{i} + (14 + 6)\mathbf{j} = -2\mathbf{i} + 20\mathbf{j}$

A1

3

(b) $\mathbf{r} = (-2\mathbf{i} + 20\mathbf{j}) \times 3 + \frac{1}{2} (2\mathbf{i} - 3\mathbf{j}) \times 9$

Using constant acceleration equation to find $\mathbf{r}$

Correct equation

M1

Correct $\mathbf{r}$

A1

$$3\mathbf{i} + 46.5\mathbf{j}$$

A1

$$r = \sqrt{3^2 + 46.5^2} = 46.6 \text{ ms}^{-1}$$

Finding magnitude
Correct magnitude

M1

5

(c) $\mathbf{v} = (2T - 2)\mathbf{i} + (20 - 3T)\mathbf{j}$

Forming equation for v

M1

$$100 = (2T - 2)^2 + (20 - 3T)^2$$

Forming equation that T must satisfy

M1

$$13T^2 - 128T + 304 = 0$$

Correct Equation

A1

$$T = \frac{128 \pm \sqrt{128^2 - 4 \times 13 \times 304}}{2 \times 13}$$

Solving for T

M1

$$= 4 \text{ or } 5.85$$

Correct solutions

A1

$$T = 4$$

Selecting T = 4

A1

6

[14]

Q26.

(a) $\mathbf{a} = \frac{1000\mathbf{i} - 5000\mathbf{j}}{2000} = \frac{t}{2}\mathbf{i} - \frac{5}{2}\mathbf{j}$

Using Newton's 2nd law

M1

Correct acceleration

A1

2

(b) $\mathbf{v} = \left(\frac{t^2}{4} + c \right) \mathbf{i} + \left(-\frac{5t}{2} + d \right) \mathbf{j}$

Integration

M1

Correct integral with or without constants

A1

$t = 0, \mathbf{v} = 6\mathbf{j} \Rightarrow c = 0, d = 6$

Evaluation of constants

m1

Correct constants and correct final answer

A1

4

$\mathbf{v} = \left(\frac{t^2}{4} \right) \mathbf{i} + \left(6 - \frac{5t}{2} \right) \mathbf{j}$

(c) $\mathbf{r} = \left(\frac{t^3}{12} + e \right) \mathbf{i} + \left(6t - \frac{5t^2}{4} + f \right) \mathbf{j}$

Integration

M1

Correct integral with or without constants

A1

$t = 0, \mathbf{r} = 0\mathbf{i} + 0\mathbf{j} \Rightarrow e = 0, f = 0$

Evaluation of constants

m1

$\mathbf{r} = \left(\frac{t^3}{12} \right) \mathbf{i} + \left(6t - \frac{5t^2}{4} \right) \mathbf{j}$

Correct constants and correct final answer

A1

4

Q27.

(a) $\mathbf{r} = (4\mathbf{i} + 7\mathbf{j})t + \frac{1}{2}(-2\mathbf{i} + 4\mathbf{j})t^2$

Finding $\mathbf{r}$

M1

$= (4t - t^2)\mathbf{i} + (7t - 2t^2)\mathbf{j}$

Correct $\mathbf{r}$

A1

2

(b) $4t - t^2 = 3$

Forming equation for one component

M1

$t^2 - 4t + 3 = 0$

$(t - 1)(t - 3)$

Solving quadratic equation

m1

$(t - 1)(t - 3)$

Solving quadratic equation

m1

$t = 1 \text{ or } t = 3$

Two solutions

A1

$7t - 2t^2 = 5$

Forming and solving quadratic equation

M1

$2t^2 - 7t + 5 = 0$

$(2t - 5)(t - 1)$

$t = 1 \text{ or } t = 2.5$

Two solutions

A1

Selecting correct solution

6

[8]

(a) (i) $\vec{AB} = (70\mathbf{i} - 40\mathbf{j}) - (30\mathbf{i} - 35\mathbf{j}) = 40\mathbf{i} - 5\mathbf{j}$
cao

1

$$(ii) \quad 40\mathbf{i} - 5\mathbf{j} = 10\mathbf{v} + \frac{1}{2} (0.2\mathbf{i} + 0.1\mathbf{j}) \times 100$$

Using their $\vec{AB}$
Correct RHS of the equation

$$10\mathbf{v} = 30\mathbf{i} - 10\mathbf{j}$$

correct v

$$\mathbf{v} = 3\mathbf{i} - \mathbf{j}$$

3

(b) $3\mathbf{i} - \mathbf{j} = \mathbf{u} + (0.2\mathbf{i} + 0.1\mathbf{j}) \times 10$
Correct RHS

$$\mathbf{u} = \mathbf{i} - 2\mathbf{j}$$

3

(c) $30\mathbf{i} - 35\mathbf{j} = (\mathbf{i} - 2\mathbf{j}) \times 10 + \frac{1}{2} (0.2\mathbf{i} + 0.1\mathbf{j}) \times 100 + \mathbf{r}_0$
 Equation of correct form for $t = 10$ or 20 with their $\mathbf{u}$
 Correct equation with their $\mathbf{u}$

M1 A1

$$30\mathbf{i} - 35\mathbf{j} = 20\mathbf{j} - 15\mathbf{j} + \mathbf{r}_0$$

$$\mathbf{r}_0 = 10\mathbf{i} - 20\mathbf{j}$$

Correct position vector

A1

3

[10]

Q29.

(a) $\mathbf{v} = -2\sin\left(\frac{t}{20}\right)\mathbf{i} + 4\cos\left(\frac{t}{20}\right)\mathbf{j}$
Differentiating
i component correct
j component correct

M1 A1 A1

3

(b) $\mathbf{v} = 4\mathbf{j}$
Correct velocity at $t = 0$

B1

Travelling north

Travelling north with $\mathbf{v} = n\mathbf{j}$ where $n > 0$

B1

2

(c) $2\sin\left(\frac{t}{20}\right) = 0$
Equation for t based on $0\mathbf{i}$

M1

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$$\frac{t}{20} = \pi$$

$$t = 20\pi$$

Correct solution

A1

2

[7]

Q30.

(a) $\mathbf{F} = (2 + 6 - 2)\mathbf{i} + (3 - 8 + 10)\mathbf{j}$
Summing the forces

M1

$$= 6\mathbf{i} + 5\mathbf{j}$$

Correct resultant

A1

2

(b) $\mathbf{a} = \frac{1}{8} (6\mathbf{i} + 5\mathbf{j}) = 0.75\mathbf{i} + 0.625\mathbf{j}$

Dividing by 8 to get correct value

B1

1

(c) (i) $\mathbf{v} = (3\mathbf{i} - 2\mathbf{j}) + 40(0.75\mathbf{i} + 0.625\mathbf{j})$

use of $\mathbf{v} = \mathbf{u} + \mathbf{at}$

M1

$= 33\mathbf{i} + 23\mathbf{j}$

Correct $\mathbf{v}$

A1

2

(ii) $\mathbf{r} = (3\mathbf{i} - 2\mathbf{j}) \times 40 + \frac{1}{2} (0.75\mathbf{i} + 0.625\mathbf{j}) \times 40^2$

Vector constant acceleration equation to find $\mathbf{r}$

Correct equation

M1 A1

$= 720\mathbf{i} + 420\mathbf{j}$

Correct position vector

A1

$r = \sqrt{720^2 + 420^2} = 834 \text{ m}$

Finding magnitude

Correct magnitude

M1 A1

5

[10]

Q31.

(a) $44\mathbf{i} + 28\mathbf{j} = 10(4\mathbf{i} + 2\mathbf{j}) + 50\mathbf{a}$

Equation to find $\mathbf{a}$ using position vector

Correct equation

M1/A1

$$\mathbf{a} = 0.08\mathbf{i} + 0.16\mathbf{j}$$

Solving for a

Correct a

M1/A1

4

$$(b) \quad \mathbf{r} = (4\mathbf{i} + 2\mathbf{j})t + \frac{1}{2}(0.08\mathbf{i} + 0.16\mathbf{j})t^2$$

Equation for r

One component correct

M1/A1

$$= (4t + 0.04t^2)\mathbf{i} + (2t + 0.08t^2)\mathbf{j}$$

Second component correct

A1

3

$$(c) \quad (i) \quad 96 = 4T + 0.04T^2 \quad 72 = 2T + 0.08T^2$$

Forms and solves equation for one component

M1

$$T = 20 \text{ or } -120 \quad T = 20 \text{ or } -45$$

Correct solutions

A1

$$\text{Hence } T = 20$$

Forms and solves equation for other component

Correct solutions plus correct conclusion

M1/A1

4

$$(d) \quad \mathbf{v} = (4\mathbf{i} + 2\mathbf{j}) + (0.08\mathbf{i} + 0.16\mathbf{j}) \times 20$$

Finding v at t = 20

M1

$$= 5.6\mathbf{i} + 5.2\mathbf{j}$$

Correct v

A1

$$v = \sqrt{5.6^2 + 5.2^2} = 7.64 \text{ ms}^{-1}$$

Finding correct speed

A1



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